

Assignment 7: Applied Deep Learning - CNN Concepts

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1. What does image classification mean? Explain with one example.

Image classification is basically when you show a computer an image and it tries to figure out what is in it. A good real-world example is Google Photos - if you take a photo of a dog and upload it, the app looks at all the pixels and says this is a dog. That is image classification. The model has been trained on many images so it can recognize what is in a new photo it has never seen before.

2. Why is using a Dense Network not always a suitable choice for images?

A dense network connects every single pixel to every single neuron, which means for even a small image you end up with millions of connections. That is a lot to train and it is very slow. Also, it does not understand that pixels next to each other are related - it just sees a flat list of numbers with no sense of space or shape. That is why it is not great for images.

3. In a CNN, what is convolution, and what does a filter or kernel do?

Convolution is when a small grid of numbers called a filter or kernel slides across the image and looks at one small patch at a time. It multiplies its values with the pixels underneath and adds them up to get one number. It keeps doing this across the whole image. Each filter is trained to detect something specific like an edge or a curve. So convolution is basically the CNN's way of scanning the image for patterns.

4. What is a feature map in a convolutional neural network?

A feature map is what you get after a filter has scanned the whole image. It is like a new, smaller image that shows where the filter found its pattern. If the filter was looking for vertical edges, the feature map will be bright where there are vertical edges and dark everywhere else. You usually have many filters running at the same time, so you get a whole stack of feature maps.

5. Why is weight sharing used in CNNs? Explain.

Weight sharing means the same filter is reused across the whole image instead of having a different one for every spot. This saves a ton of parameters - you only need to learn the filter once. It also means if the model learns to detect something in the top-left corner of an image, it can detect that same thing anywhere else in the image without needing to relearn it.

6. How does the ReLU activation function work? Write the ReLU output for [-3, 0, 2, 5].

ReLU is really simple - if the number is positive, keep it. If it is zero or negative, turn it to zero.

For the values [-3, 0, 2, 5]:

-3 becomes 0 (negative, so it gets wiped)

0 stays 0

2 stays 2 (positive, no change)

5 stays 5 (positive, no change)

Output: [0, 0, 2, 5]

7. What is max pooling, and why is it used in CNNs?

Max pooling shrinks the feature map by going through it in small windows like 2x2 and just keeping the biggest number in each window. So a 4x4 feature map becomes a 2x2 one. It is used to make the data smaller so the model trains faster. It also makes the model less sensitive to tiny shifts in the image - if a feature moves one pixel, the max value usually stays the same.

8. In the Fashion-MNIST code, why are images reshaped from 28x28 to 28x28x1?

CNN layers need to know how many colour channels the image has. A colour image has 3 channels (red, green, blue). Fashion-MNIST images are grayscale so they only have 1 channel, but we still need to tell the model that. Reshaping from 28x28 to 28x28x1 just adds that extra 1 so the model knows it is dealing with a single-channel image. Without it, the model would throw an error.

9. What is the difference between training data, validation data, and test data?

Training data is what the model actually learns from - it sees these examples over and over and adjusts its weights. Validation data is used while training to check how the model is doing on examples it has not learned from yet. I use it to decide things like whether to stop training early or change some settings. Test data is only used at the very end, after everything is done, to get a final honest score on how well the model actually works.

10. What is overfitting? If training accuracy is high but validation accuracy is low, what does this indicate?

Overfitting is when a model basically memorizes the training data instead of learning general patterns. It does great on what it has seen before but fails when given new data. So if training accuracy is high but validation accuracy is low, that is a sign the model is overfitting - it is not really learning, just remembering. To fix it you can add dropout, simplify the model, or use more training data.